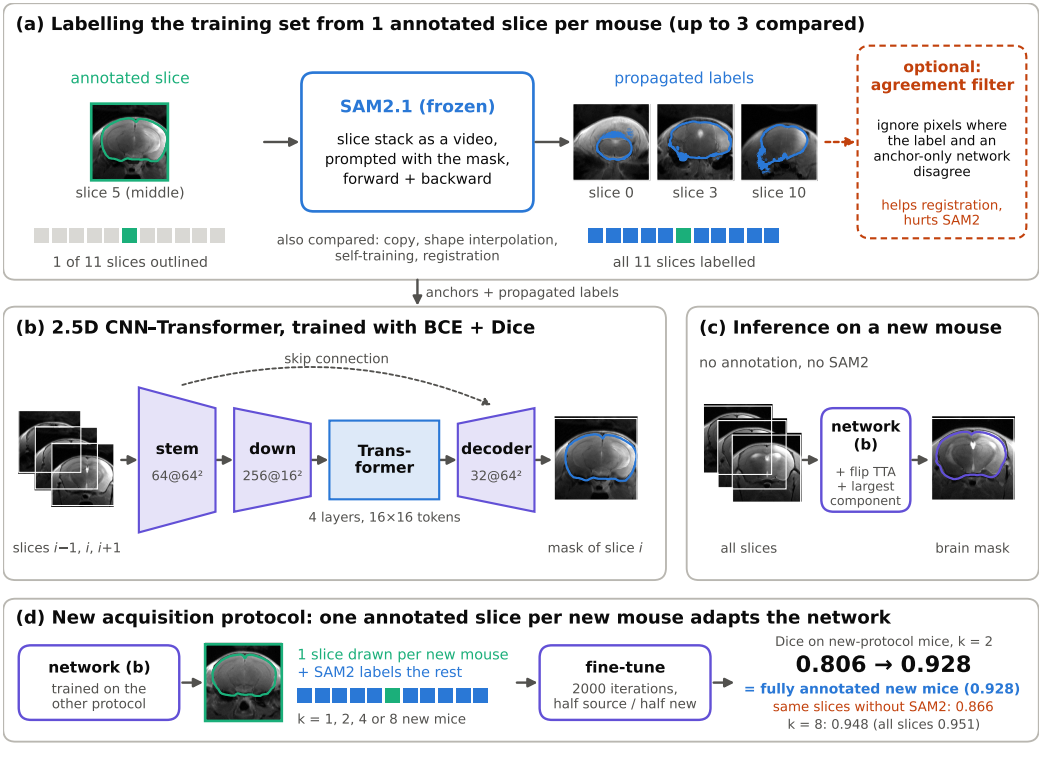

(a) Labelling the training set from 1 annotated slice per mouse (up to 3 compared)
annotated slice
slice 5 (middle)
1 of 11 slices outlined
SAM2.1 (frozen)
slice stack as a video, prompted with the mask, forward + backward
also compared: copy, shape interpolation, self-training, registration
propagated labels
slice 0
slice 3
slice 10
all 11 slices labelled
optional: agreement filter
ignore pixels where the label and an anchor-only network disagree
helps registration, hurts SAM2
anchors + propagated labels
(b) 2.5D CNN–Transformer, trained with BCE + Dice
skip connection
stem
64@64²
down
256@16²
Trans-former
4 layers, 16×16 tokens
decoder
32@64²
slices i−1, i, i+1
mask of slice i
(c) Inference on a new mouse
no annotation, no SAM2
network (b)
+ flip TTA
+ largest component
all slices
brain mask
(d) New acquisition protocol: one annotated slice per new mouse adapts the network
network (b)
trained on the other protocol
1 slice drawn per new mouse
+ SAM2 labels the rest
k = 1, 2, 4 or 8 new mice
fine-tune
2000 iterations, half source / half new
Dice on new-protocol mice, k = 2
0.806 → 0.928
= fully annotated new mice (0.928)
same slices without SAM2: 0.866
k = 8: 0.948 (all slices 0.951)